PORTFOLIO 5 – Distributed and Cloud Database Exercise

(1) Cloud Database Deployment

The University Student Management Database in this exercise was implemented in the Cloud using MongoDB Atlas. Also, I used MongoDB Compass to remotely connect to the MongoDB Atlas Cluster and this enabled the Database Records to be seen and managed. The screenshots I have uploaded provide proof of the Cloud Deployment and Remote Access to the Database.

(2) Replication

Replication enables copies of Database Data to be maintained across Multiple Database Servers. Considering MongoDB, this Cloud Database Product uses Replica Sets to deliver Data Redundancy, and enable Recovery in the event that a Database Server becomes Unavailable. Also, the MongoDB Atlas solution is a Managed Cloud Infrastructure, and Failover Testing with Manual Replica-set Configuration wasn't performed in this exercise.

(3) Sharding

This is a method of distributing a large Database across Multiple Servers. It enables increasing amounts of data and workloads to be efficiently handled by a Database by distributing Storage and Processing.
As an example, a University can grow to have a very large student population and at that stage, the Database could then be distributed across multiple Database Servers using a shard key.

(4) Availability

Having a Cloud Database solution like MongoDB Atlas provides Remote Access to Database records but this is largely hinged on Service and Network Connection availability.

(5) Consistency

Database operations must be valid and the Data reliable. This is what consistency ensures, and MongDB is designed with consistency mechanisms in Database Operations and Replica-set configuration.

(6) Final Thoughts

This implementation did show a MongoDB Database deployed in the Cloud and Remotely accessed using MongoDB Compass. It brought to light the concepts of Replication, Sharding, Availability, Consistency and how to make the most of them in a Distributed Database System.